

A MULTIPLE COMPARISON OF RANK PROCEDURE FOR A ONE-WAY ANALYSIS OF VARIANCE

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1. Introduction and Summary

Multiple comparison rank procedures for the simultaneous comparison of means in a one-way analysis of variance classification have been put forward by various authors. In this paper the attention is mainly restricted to the Wilcoxon-type rank analogues to Scheffé's (1953, 1959) parametric procedure. Nemenyi (1963) proposed such a procedure based on the Kruskal-Wallis (1952) k -sample extension of Wilcoxon's test. This procedure, cf. Miller (1966) p. 168, has the unpleasant property that the outcome of the comparison between two populations depends on the observations in the other populations, and in this paper an alternative procedure is proposed which does not have this property. This procedure is similar to a procedure proposed by Dunn (1964) using the Bonferroni inequality. A comparison of the proposed procedure and Nemenyi's procedure shows that the latter can be expected to be more sensitive for comparing two populations in respect of location when the alternative is such that the observations in the other populations tend to fall between the observations in these two populations. Similarly the former is expected to be more sensitive when the alternative is such that the observations in the other populations tend to fall outside the observations in these two populations.

In Section 5 the problem of application when the sample sizes are small is briefly dealt with. The main conclusion in this respect is that for small sample sizes the critical values for the proposed procedure based on the χ^2 -approximation tend to be too large and this implies that the procedure tends to be conservative.

2. General Theoretical Results

Let $X_{11}, \dots, X_{1n_1}; \dots; X_{k1}, \dots, X_{kn_k}$ be k random and independent samples from populations with continuous cumulative distribution functions $F_1, \dots, F_k$. The null hypothesis to be tested is

$$H_0: F_1 = F_2 = \dots = F_k$$

while the alternative is that at least two populations differ in location, i.e.

$$H_1: F_i(x) = F(x + \theta_i), i, j = 1, \dots, k, \theta_i \neq \theta_j \text{ for at least one pair } i, j.$$

Let $w(x,y)$ be a non-degenerate real valued function of x and y such that $w(x,y) = -w(y,x)$ and define

$$T_{ij}(w) = \frac{1}{n_i n_j} \sum_{r=1}^{n_i} \sum_{s=1}^{n_j} w(X_{ir}, X_{js}).$$

The class of functions of the form w is denoted by W . Furthermore let σ_w^2 be the covariance under H_0 between $w(X_{ir}, X_{js_1})$ and $w(X_{ir}, X_{js_2})$ with $1 \leq s_1 < s_2 \leq n_j$. It is assumed that σ_w^2 is finite.

Finally let $I(w, K)$ denote the set of inequalities

$$-K < \frac{\sum_{i < j} a_{ij}(n) T_{ij}(w)}{\sqrt{\text{var}_0 \sum_{i < j} a_{ij}(n) T_{ij}(w)}} < K$$

where K is a positive constant, the a_{ij} are arbitrary finite real functions of the n_j such that $\lim a_{ij}(n) = a_{ij} < \infty$ for all i, j and $\lim \sqrt{N} \sum a_{ij}(n) T_{ij}(w)$ is non-degenerate where $N = \sum n_i$.

In what follows attention is restricted to large samples and in this respect it is assumed that the limit approach is such that $\lim n_i/N$ exists and is bounded away from zero for all i , where $N = \sum n_i$.

From the theory of generalized U statistics (cf. Hoefding (1948), Lehmann (1951) and Crouse (1960)) it readily follows that under H_0

$$E_0 T_{ij}(w) = 0$$

$$\text{var}_0 T_{ij}(w) = \frac{(n_i + n_j)}{n_i n_j} \sigma_w^2 + O\left(\frac{1}{n_i n_j}\right)$$

$$\text{covar}_0 T_{ij_1}(w) T_{ij_2}(w) = \frac{1}{n_i} \sigma_w^2 \text{ for } j_1 \neq j_2,$$

and that the limiting distribution of the random vector

$$\frac{T_{ij}(w)}{\sigma_w \sqrt{\frac{(n_i + n_j)}{n_i n_j}}}; \quad 1 \leq i < j \leq k$$

is normal with covariance matrix independent of w for all $w \in W$.

The following statement can now be made:

Statement 1. Under H_0 a limiting probability statement with respect to the simultaneous occurrence of all the inequalities of the set $I(w, K)$ which is true for at least one $w \in W$ is also true for any $w \in W$.

Let

$$X_i = \frac{1}{n_i} \sum_{j=1}^{n_j} X_{ij}, \quad i = 1, \dots, k,$$

then in the limit the Scheffé procedure for establishing which contrasts of the

form $\sum_{i < j} C_{ij}(n)/(\theta_i - \theta_j)$ (the equivalent of $\sum C_{ij}(n)/\theta_i$ with $\sum C_{ij}(n) = 0$, $\lim C_{ij}(n) = C_i < \infty$ for all i and bounded away from zero for at least one i) are significantly different from zero utilizes the following probability statement.

Statement 2. Under H_0 and the assumption that the variance of the underlying distribution is finite, the limiting probability is $(1-\alpha)$ that all inequalities of the form

$$-\chi_{\alpha, k-1} \leq \frac{\sum C_{ij}(n)/(X_i - X_j)}{\sqrt{\text{var}_0 \sum C_{ij}(n)/(X_i - X_j)}} \leq \chi_{\alpha, k-1}$$

are simultaneously satisfied.

Let $w_1(x, y) = x - y$, then

$$(X_i - X_j) = T_{ij}(w_1)$$

and since $w_1 \in W$ it follows from Statements 1 and 2 that:

Statement 3. Under H_0 the limiting probability is $(1-\alpha)$ that all inequalities of the form

$$-\chi_{\alpha, k-1} \leq \frac{\sum C_{ij}(n) T_{ij}(w)}{\sqrt{\text{var}_0 \sum C_{ij}(n) T_{ij}(w)}} \leq \chi_{\alpha, k-1}$$

are simultaneously satisfied for any $w \in W$.

3. A Wilcoxon-type procedure

Let

$$w_2(x, y) = \frac{1}{2} \text{sign}(x - y)$$

then $V_{ij} = n_i n_j T_{ij}(w_2)$ is the two-sample Wilcoxon statistic between $X_{i1}, \dots, X_{in_i}$ and $X_{j1}, \dots, X_{jn_j}$ standardized with respect to its mean. Since $w_2 \in W$ one has by utilizing Statement 3, an alternative rank analog to Scheffé's procedure for the simultaneous comparison of means in a one-way analysis of variance for large sample sizes. (By similar reasoning, cf. Crouse (1967), it can be shown that for large sample sizes a Wilcoxon-type rank analog of the classical F test is to accept H_0 when

$$TV = \frac{12}{N + \frac{k}{2}} \sum_{i < j} \frac{V_{ij}^2}{n_i n_j} < \chi_{\alpha, k-1}^2$$

and in practice one could firstly apply this test and if H_0 is rejected perform the above procedure.)

For simple paired comparisons the interpretation of the proposed procedure

is straightforward namely for every $i < j$ one of the hypotheses $F_i > F_j$, $F_i = F_j$, $F_i < F_j$ is accepted depending on whether the observed value of $V_{ij}/\sqrt{\text{var}_0 V_{ij}}$, i.e. the observed two-sample Wilcoxon statistic standardized under H_0 , is contained in $(-\infty, -\chi_{\alpha, k-1})$, $(-\chi_{\alpha, k-1}, \chi_{\alpha, k-1})$, $(\chi_{\alpha, k-1}, \infty)$. Clearly the comparison of any two populations only depends on the observations in these populations and hence this procedure does not suffer from the weakness of the procedure proposed by Nemenyi (1963).

The interpretation of the procedure in general is not quite as obvious as for simple paired comparisons, the question being: which contrasts are of general practical importance? The same question applies to Scheffé's parametric procedure and one way of looking at this problem is the following: With Scheffé's procedure all contrasts can be tested simultaneously while with Tukey's procedure (cf. e.g. Scheffé (1959), and Miller (1965)) attention is restricted to simple paired comparisons in respect of location. On the one hand one is never interested in testing for all contrasts, but on the other hand situations do arise where one desires a more conclusive answer than that provided by only comparing populations pairwise in respect of location. The ideal situation appears to be somewhere between these two extremes. In most practical one-way classification problems the emphasis, to this author, is on pairwise comparisons of populations and on pairwise comparisons of groups of populations, the populations within each group assumed to be identical. For instance in this type of problem the data often suggests that $F_{a_1} = \dots = F_{a_{k_1}} = G_1$ and $F_{b_1} = \dots = F_{b_{k_2}} = G_2$, i.e. $F_{a_1}, \dots, F_{a_{k_1}}$ constitute one group and $F_{b_1}, \dots, F_{b_{k_2}}$ another, and one then wishes to test $G_1 = G_2$ against the alternative that G_1 and G_2 differ in respect of location. The appropriate parametric test for this problem is essentially based on

$$(1) \quad \frac{\sum_{i=1}^{k_1} n_{a_i} X_{a_i}}{\sum_{i=1}^{k_1} n_{a_i}} - \frac{\sum_{j=1}^{k_2} n_{b_j} X_{b_j}}{\sum_{j=1}^{k_2} n_{b_j}},$$

i.e. the difference between the means of the two groups. Alternatively this is the test for testing

$$\frac{\sum_{i=1}^{k_1} n_{a_i} \theta_{a_i}}{\sum_{i=1}^{k_1} n_{a_i}} - \frac{\sum_{j=1}^{k_2} n_{b_j} \theta_{b_j}}{\sum_{j=1}^{k_2} n_{b_j}} = 0$$

and from this point of view the contrasts of practical importance are all contrasts of the above form with $\{a_i; i = 1, \dots, k_1\}$ and $\{b_j; j = 1, \dots, k_2\}$ two nonempty mutually exclusive subsets of $\{1, 2, \dots, k\}$. The total number of contrasts of this form is

$$\frac{1}{2} \binom{k}{1}(k-1) + \binom{k}{2}(k-2) + \binom{k}{3}(k-3) + \dots$$

$$\begin{aligned}
 & + \frac{1}{2} \binom{k}{2} \binom{k-2}{2} + \binom{k}{3} \binom{k-3}{2} + \binom{k}{4} \binom{k-4}{2} + \dots \\
 & + \frac{1}{2} \binom{k}{3} \binom{k-3}{3} + \binom{k}{4} \binom{k-4}{3} + \binom{k}{5} \binom{k-5}{3} + \dots \\
 & + \dots
 \end{aligned}$$

The statistic (1) can be written as

$$\sum_{i=1}^{k_1} \sum_{j=1}^{k_2} C_{ij}(n) (X_{ai} - X_{bj})$$

where

$$C_{ij}(n) = \frac{n_{ai} \cdot n_{bj}}{\sum_{i=1}^{k_1} n_{ai} \sum_{j=1}^{k_2} n_{bj}}$$

and according to Scheffé's procedure, for large samples and assumptions of normality, one of the hypotheses $G_1 > G_2$, $G_1 = G_2$, $G_1 < G_2$ is accepted depending on whether the observed value of

$$\begin{aligned}
 & \frac{\sum_{i=1}^{k_1} \sum_{j=1}^{k_2} C_{ij}(n) (X_{ai} - X_{bj})}{\sqrt{\text{var}_0 \sum_{i=1}^{k_1} \sum_{j=1}^{k_2} C_{ij}(n) (X_{ai} - X_{bj})}} = \\
 & \frac{\sum_{i=1}^{k_1} \sum_{j=1}^{k_2} n_{ai} n_{bj} (X_{ai} - X_{bj})}{\sqrt{\text{var}_0 \sum_{i=1}^{k_1} \sum_{j=1}^{k_2} n_{ai} n_{bj} (X_{ai} - X_{bj})}}
 \end{aligned}$$

is contained in $(-\infty, -\chi_{\alpha, k-1})$, $(-\chi_{\alpha, k-1}, \chi_{\alpha, k-1})$, $(\chi_{\alpha, k-1}, \infty)$. Likewise for the proposed rank procedure one of these hypotheses is accepted depending on the interval in which

$$\frac{\sum_{i=1}^{k_1} \sum_{j=1}^{k_2} V_{ai} b_j}{\sqrt{\text{var}_0 \sum_{i=1}^{k_1} \sum_{j=1}^{k_2} V_{ai} b_j}}$$

is contained. Clearly the statistic

$$\sum_{i=1}^{k_1} \sum_{j=1}^{k_2} V_{ai} b_j$$

is the Wilcoxon statistic, standardized with respect to its mean, between the

$$\sum_{i=1}^{k_1} n_{ai}$$

observations of the one group and the

$$\sum_{j=1}^{k_2} n b_j$$

observations of the other.

4. Comparison with Nemenyi's procedure

Although the property of Nemenyi's procedure, that the outcome of a comparison between two populations depends on the observations in the other populations, is regarded as undesirable it is not obvious what the influence of this dependence is. To throw light on this matter consider Nemenyi's procedure for comparing F_i with F_j , i.e. one of the hypotheses $F_i > F_j$, $F_i = F_j$, $F_i < F_j$ is accepted depending on whether the observed value of

$$\frac{R_i - R_j}{\sqrt{\text{var}_0(R_i - R_j)}}$$

is contained in $(-\infty, -\chi_{\alpha, k-1}), (-\chi_{\alpha, k-1}, \chi_{\alpha, k-1}), (\chi_{\alpha, k-1}, \infty)$ where R_i is the mean of the ranks of the X_i -observations in the combined sequence of N observations. Alternatively for the proposed procedure one of these hypotheses is accepted depending on the interval in which

$$\frac{V_{ij}}{\sqrt{\text{var}_0 V_{ij}}}$$

is contained.

It can readily be shown that

$$n_i R_i - \frac{1}{2} n_i (N+1) = \sum_{\substack{j=1 \\ (j \neq i)}}^k V_{ij},$$

$$R_i - R_j = \frac{N V_{ij}}{n_i n_j} + \frac{1}{n_i n_j} Q,$$

where

$$Q = \sum_{h=1}^k \left(n_j V_{ih} + n_i V_{hj} + n_h V_{ji} \right) \quad (h \neq i, j)$$

and

$$(2) \quad \begin{aligned} & E(n_j V_{ih} + n_i V_{hj} + n_h V_{ji}) \\ &= n_h n_i n_j \left(\frac{1}{2} - \int F_j dF_i - \int F_i dF_h + \int F_j dF_h \right). \end{aligned}$$

In broad terms the influence of the distribution F_h , through its presence in the term Q , on the test based on $R_i - R_j$ relative to the test based on V_{ij} can be summarized as follows:

Firstly if $F_i = F_j$ with F_h identical to F_i (or F_j), then expression (2) vanishes and hence in terms of expected values the distribution F_h , through its presence in the

term Q , does not contribute to $R_i - R_j$, as a measure of the departure between F_i and F_j .

Secondly if $F_h < F_i < F_j$ and $\int F_j dF_i + \int F_i dF_h > 1.5$ then expression (2) and $E(V_{ij})$ have opposite signs, thus indicating that the distribution F_h , through its presence in the term Q , has an adverse effect on the power of the test based on $R_i - R_j$, relative to the test based on V_{ij} . The same applies to the corresponding cases $F_h > F_i > F_j$, $F_i < F_j < F_h$ and $F_i > F_j > F_h$.

Thirdly if $F_i < F_h < F_j$ and $\int F_j dF_h - \int F_i dF_h > \frac{1}{2}$ then expression (2) and $E(V_{ij})$ have the same sign and hence the distribution F_h , through its presence in the term Q , will tend to increase the power of the test based on $R_i - R_j$, relative to the test based on V_{ij} . The same applies to the corresponding case $F_i > F_h > F_j$.

For other choices of F_h , relative to the alternative $F_i \neq F_j$, the influence of F_h on the test based on $R_i - R_j$, relative to the test based on V_{ij} is not as obvious and conclusive as above.

In a similar fashion every distribution F_h , for all $h \neq i, j$, has an influence on the power of the test based on $R_i - R_j$, relative to the test based on V_{ij} and generally speaking this influence can be formulated as follows: When comparing two populations in respect of location and the alternative is such that the other populations tend to separate these two populations, in the sense that the observations tend to fall between the observations in these two populations, then Nemenyi's procedure can be expected to be more sensitive than the proposed procedure. On the other hand if the observations in the other populations have a tendency to fall outside the region of the observations in these two populations the opposite can be expected to be true.

5. Practical Considerations

Approximate critical values for the proposed procedure can readily be deduced from the corresponding critical values of the χ^2 distribution. These critical values, denoted by $V_{\alpha; k; m, n}$, are given in Table 1, the interpretation being as follows: Suppose that in a one-way classification with k samples one wishes to compare one group of samples with a total of m observations with a second group with a total of n observations. According to the above procedure the statistic V , i.e. the Wilcoxon statistic standardized with respect to its mean, is calculated for these two groups. The null hypothesis is accepted if

$$-V_{\alpha; k; m, n} < V < V_{\alpha; k; m, n}$$

and asymptotically under the null hypothesis the probability is just in excess of $(1-\alpha)$ that all inequalities of the above form are simultaneously satisfied. Similarly when V is outside the above interval the mean of the one group is regarded as being above or below the mean of the other, depending on the sign of V .

To determine how satisfactory the approximate critical values are for small sample sizes two comparisons are made.

Firstly in the parametric case, with the common population variance known, the probability error of every individual decision is $2P(Z \geq \chi_{\alpha, k-1})$ where Z is a $N(0,1)$ variate. Similarly one can now, for small sample sizes, determine the

critical values for the proposed procedure such that the probability error of each individual decision is at most $2P(Z > \chi_{a,k-1})$. For $k = 3, 4$; $a = 0.10, 0.05, 0.01$ and the number of observations in each group not exceeding 10 these critical values denoted by $V_{a;k;m,n}^*$ are given in Table 2. These critical values are derived from the probability distribution for the Wilcoxon statistic given by Owen (1962). A comparison of these critical values with those given in Table 1 shows that at the 10% level of significance the correspondence is remarkably good while at the 5% and 1% levels the χ^2 values tend to be larger and thus are conservative.

Secondly in Table 3 the exact probability errors, for $k = 3$ and sample sizes not exceeding four, are given for the procedure based on the approximate χ^2 values. Asymptotically these probability errors will correspondingly be just below 10% and 5%, and the calculated probability errors are thus generally conservative. This is to be expected because due to the discreteness of the Wilcoxon statistic not all of the comparisons can, for small sample sizes, result in significant differences, e.g. for $k = 3$, $n_1 = 2$, $n_2 = 3$, $n_3 = 4$ not one of the hypotheses $F_1 = F_2$, $F_1 = F_3$, $F_2 = F_3$ can be rejected.

The general conclusion is thus that the critical values of Table 1 can safely be used.

In Table 4 limited critical values of the test based on T_V , i.e. a Wilcoxon-type rank analog of the F test, are given. Although much more work can still be done in this respect it is conjectured that the χ^2 approximation can safely be used at the 5% and 10% levels of significance when the individual sample sizes exceed three and do not differ too widely from each other.

To illustrate the procedure consider the following example. The following are four random and independent samples from rectangular distributions with the same range but with mean values 0.8, 0.9, 0.4 and 1.2 respectively.

I	II	III	IV
0.442	1.077	0.602	0.945
1.262	1.315	0.037	0.907
0.894	0.654	0.495	1.101
1.247	1.355	0.623	1.467
0.645		0.788	1.521
0.828			1.110
0.485			

The problem is to perform a multiple comparison test of means and groups of means at the 10% level of significance.

By writing the observations in each sample in decreasing order it is easy to count the number of times an observation from sample i exceeds an observation from sample j , say U_{ij} , and V_{ij} then readily follows from the relation $V_{ij} = U_{ij} - \frac{1}{2}n_i n_j$. The values of V_{ij} are

	II	III	IV
I	-8	8.5	-13
II		9	-2
III			-15

and the value of the corresponding rank equivalent test statistic,

$$TV = \frac{12}{N + \frac{k}{2}} \sum_{i < j} \frac{V_{ij}^2}{n_i n_j},$$

of the F-test is now found to be 10.1494. The corresponding critical value is $\chi_{0.10,3}^2 = 6.251$ and the hypothesis of no difference is rejected at the 10% level of significance.

To determine whether there is a significant difference between any two means the above values of V_{ij} are compared with the critical values in Table 1 at the 10% level of significance. The only observed value found to be significant is $V_{III IV}$ and the conclusion is that the population mean of sample III is at the 10% level of significance lower than that of sample IV.

From the above values of V_{ij} the samples can be placed in the following increasing order,

III, I, II, IV

with II and IV close to each other. This suggests that II and IV may form a group and a comparison of III with II + IV results in a value of

$$\begin{aligned} V_{III II} + V_{III IV} &= -9 - 15 \\ &= -24 \end{aligned}$$

The critical value in this case is $V_{0.10;4;5,10} = 21$ and the observed value is thus significant. Next one could also compare I with II + IV, i.e.

$$V_{I II} + V_{I IV} = -21,$$

which is not significant, and I + III with II + IV, i.e.

$$V_{I II} + V_{I IV} + V_{III II} + V_{III IV} = -45$$

which is significant, the 5% critical value being 38, and the logical conclusion is that these four samples are representative of at least two different populations, i.e. II and IV from one population and I and III from another with the mean of the former the highest.

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REFERENCES

- CROUSE, C.F. (1960). Unpublished Ph.D. thesis, University of London.
- CROUSE, C.F. (1967). A class of distribution-free analysis of variance tests, *S.Afr.Statist. J.* (1967), 1, 75-80.
- DUNN, O.J. (1964). Multiple comparisons using rank sums. *Technometrics*, 6, 241-252.

TABLE 1
Values of $V_{a;k;m,n}$

k	m	n	a		
			0.01	0.05	0.10
3	4	4	-	-	8
		5	-	-	9
	5	4	-	-	9
		5	-	12.5	10.5
	6	3	-	-	9
		4	-	12	11
		5	-	14	12
		6	-	16	14
	7	3	-	-	9.5
		4	-	13	12
		5	-	15.5	13.5
		6	-	18	16
		7	24.5	19.5	17.5
	8	3	-	12	11
		4	-	15	13
		5	-	17	15
		6	24	19	17
		7	27	22	19
		8	29	24	21
	9	3	-	13.5	12.5
		4	-	16	14
		5	-	18.5	16.5
		6	26	21	19
		7	29.5	23.5	20.5
		8	32	26	23
		9	34.5	28.5	24.5
	10	2	-	-	10
		3	-	15	13
		4	-	18	16
		5	25	20	18
		6	28	23	20
		7	32	26	22
		8	35	28	25
		9	38	30	27
		10	41	33	29

HOEFFDING, W. (1948). A class of statistics with asymptotically normal distribution. *Ann. Math. Statist.*, **19**, 293-325.

KRUSKAL, W.H. and WALLIS, W.A. (1952). Use of ranks in one-criterion variance analysis. *J. Am. Statist. Assoc.*, **47**, 583-621.

TABLE 1 (*Continued*)

k	m	n	α		
			0.01	0.05	0.10
4	5	5	-	-	12.5
		6	-	-	12
		5	-	-	14
		6	-	18	16
	7	4	-	-	14
		5	-	17.5	15.5
		6	-	20	18
		7	-	22.5	20.5
	8	3	-	-	-
		4	-	17	15
		5	-	20	18
		6	-	22	20
		7	-	25	22
		8	-	27	24
	9	3	-	-	-
		4	-	-	17
		5	-	21.5	19.5
		6	-	24	22
		7	-	26.5	24.5
		8	35	30	26
		9	38.5	32.5	28.5
	10	3	-	-	15
		4	-	20	18
		5	-	23	21
		6	-	26	24
		7	35	29	26
		8	38	32	29
		9	42	35	31
		10	45	37	34

LEHMANN, E.L. (1951). Consistency and unbiasedness of certain nonparametric tests, *Ann. Math. Statist.*, 22, 165-179.

MILLER, R.G. (1966). *Simultaneous statistical inference*. McGraw-Hill Book Company.

NEMENYI, P. (1963). Unpublished Ph.D. thesis, Princeton University

OWEN, D.B. (1962). *Handbook of statistical tables*, Addison-Wesley Publishing Company.

SCHEFFÉ, H. (1953). A method for judging all contrasts in the analysis of variance, *Biometrika*, 40, 87-104.

TABLE 2
Values of $V_{a;k;m,n}^*$

k	m	n	α		
			0.01	0.05	0.10
3	4	4	-	-	8
		5	-	-	9
		5	-	12.5	10.5
	6	3	-	-	9
		4	-	12	11
		5	-	14	12
		6	18	16	14
	7	3	-	-	10.5
		4	-	13	12
		5	-	15.5	13.5
		6	20	17	16
		7	22.5	19.5	17.5
	8	3	-	12	11
		4	-	15	13
		5	20	17	15
		6	23	19	17
		7	25	21	19
		8	28	24	21
	9	3	-	13.5	12.5
		4	-	16	15
		5	21.5	18.5	16.5
		6	25	21	19
		7	27.5	23.5	20.5
		8	31	26	23
		9	33.5	27.5	24.5
	10	2	-	-	10
		3	-	14	13
		4	20	17	16
		5	24	20	18
		6	27	23	20
		7	30	25	23
		8	33	28	25
		9	36	30	27
		10	39	33	29

SCHEFFÉ, H. (1959). *The analysis of variance*, John Wiley & Sons.

TABLE 2 (Continued)

k	m	n	<i>a</i>		
			0.01	0.05	0.10
4	5	5	-	-	12.5
		6	-	-	12
	6	4	-	-	12
		5	-	15	14
		6	-	17	16
	7	4	-	-	13
		5	-	16.5	15.5
		6	-	19	18
		7	24.5	21.5	19.5
	8	3	-	-	12
		4	-	16	15
		5	-	19	17
		6	24	21	20
		7	27	24	22
		8	30	26	24
	9	3	-	-	13.5
		4	-	18	16
		5	-	20.5	18.5
		6	27	23	21
		7	29.5	26.5	23.5
		8	33	29	26
		9	36.5	31.5	28.5
	10	3	-	-	15
		4	-	19	18
		5	25	22	21
		6	29	25	23
		7	32	28	26
		8	36	31	28
		9	39	34	31
		10	42	37	33

TABLE 3

Exact probability errors for the critical values given in Table 1 for $k = 3$ and sample sizes not exceeding four

Sample sizes			α	
n_1	n_2	n_3	0.05	0.10
4	2	2	-	0.0343
4	3	1	-	0.0286
3	3	3	-	0.0679
4	3	2	-	0.0524
4	4	1	-	0.0540
4	3	3	0.0095	0.0800
4	4	2	0.0184	0.0412
4	4	3	0.0350	0.0743
4	4	4	0.0235	0.0977

TABLE 4

Values of $T_{\alpha,V}$ where $P(T_V \geq T_{\alpha,V}) \leq \alpha$ for $k = 3$ and sample sizes not exceeding four

Sample sizes			α		
n_1	n_2	n_3	0.01	0.05	0.10
2	2	2			4.800
3	2	1	-	-	4.400
3	3	1	-	5.294	4.353
3	2	2	-	4.588	4.471
4	2	1	-	-	4.235
3	3	2	-	5.509	4.456
4	2	2	-	5.211	4.263
4	3	1	-	5.053	4.000
3	3	3	7.714	5.429	4.921
4	3	2	6.476	5.286	4.333
4	4	1	6.857	4.929	4.071
4	3	3	6.870	5.565	4.551
4	4	2	7.370	5.283	4.370
4	4	3	7.100	5.440	4.480
4	4	4	7.556	5.556	4.556
∞	∞	∞	9.210	5.991	4.605